XYZ+ST: A Unified Evaluation Framework for AI Governance State and Response Quality

XYZ+ST：AI ガバナンス状態と AI 応答品質の統合評価フレームワーク

Kimura, M. (きむら博士)
Independent Researcher | MoCKA Project
AI Co-Author: Claude (Anthropic)
Technical Note v1.0 | 2026-04-11
DOI: 10.5281/zenodo.19503666 (MoCKA preprint series)

Abstract

We present XYZ+ST, a five-dimensional unified evaluation framework for AI governance systems. The framework integrates two distinct evaluation layers: the XYZ fluid coordinate model for institutional governance state assessment, and the ST axes for AI response quality measurement. XYZ captures the dynamic trajectory of an AI governance institution across three dimensions: X (Protocol Fidelity), Y (Record Quality), and Z (Governance Stability). ST extends this with S (Strength: elimination of redundancy) and T (Terminal-Only: single-stage execution discipline). Empirical validation using the MoCKA multi-agent orchestration system demonstrates a ΔZYXTS total of +0.6367 between uncontrolled (total=0.2458) and essence-injected conditions (total=0.8825), confirming the framework's sensitivity to governance intervention.

Keywords: AI governance, evaluation framework, fluid coordinates, response quality, multi-agent systems, MoCKA

1. Introduction

The evaluation of AI governance systems faces a fundamental challenge: existing metrics either assess system-level behavior (incident rates, compliance scores) or response-level quality (accuracy, fluency), but rarely both in a unified framework. This bifurcation prevents practitioners from understanding how institutional governance structure influences individual AI response quality.

We propose XYZ+ST as a solution to this problem. The framework operates at two levels simultaneously:

XYZ (Governance State Layer): Three-dimensional coordinate system tracking institutional health as a continuous trajectory P(t) = (X(t), Y(t), Z(t)) rather than a binary compliant/non-compliant judgment.

ST (Response Quality Layer): Two-dimensional scoring axes applied to individual AI responses, measuring execution discipline (T) and communicative efficiency (S).

Together, XYZ+ST provides a complete picture: the institutional context in which AI operates (XYZ) and the quality of individual outputs within that context (ST).

2. Framework Definition

2.1 The XYZ Governance State Model

Institutional governance health is represented as a three-dimensional coordinate:

$$P(t) = (X(t), Y(t), Z(t))$$

Axis	Name	Definition	Empirical Value (MoCKA)
X	Protocol Fidelity	Degree to which AI agents follow institutional rules without deviation	0.632 (weakest axis)
Y	Record Quality	Completeness and accuracy of 5W1H structured event records	0.826
Z	Governance Stability	Consistency of governance enforcement over time	0.819 (↑from 0.604)

The Z-axis exhibits a diagnostic technology effect: apparent decline in Z following MoCKA deployment reflects improved detection of previously invisible governance failures, not actual degradation. Z=0.604→0.819 after intervention corrects a Gemini data fabrication incident (2026-04-10).

2.2 The ST Response Quality Axes

ST axes evaluate the quality of individual AI responses within the governance context:

Axis	Name	Scoring Criterion	Detection Method
S	Strength	Elimination of redundant language, hedging, and preamble	Vague term count: 'perhaps', 'generally', 'may'
T	Terminal-Only	Structured step-by-step execution reaching a definitive conclusion	Presence of numbered procedural steps (1-N)

2.3 Unified ZYXTS Scoring

The five axes are scored on [0.0, 1.0] and aggregated:

```
ZYXTS_total = (Z + Y + X + T + S) / 5
```

Note: The ordering ZYXTS (rather than XYZST) reflects empirical priority: Z and Y are the most stable governance dimensions, while X, T, S are more sensitive to intervention.

3. Empirical Validation

3.1 Experimental Design

A controlled within-session experiment evaluated the effect of essence injection on ZYXTS scores. Two conditions were compared:

Condition A (baseline): Response generated without MoCKA essence injection. Simulates standard LLM central tendency behavior.

Condition B (treatment): Response generated with 230-item essence corpus injection from real MoCKA operational history (real_essence_from_raw.json, 6.3MB).

Test question: 'In MoCKA router.save processing, what must be done to guarantee recording accuracy? Provide concrete steps.'

3.2 Results

Axis	Condition A (baseline)	Condition B (essence)	Delta
Z (Depth/Specificity)	0.1667	1.0000	+0.8333
Y (Reload Verification)	1.0000	1.0000	0.0000
X (Precision)	0.0000	1.0000	+1.0000
T (Terminal-Only)	0.0000	1.0000	+1.0000
S (Strength)	0.0625	0.4125	+0.3500
TOTAL	0.2458	0.8825	+0.6367

The Y-axis showed no differential (both conditions scored 1.0), suggesting re-verification language appears even in uncontrolled responses for this question type. Future experiments should use questions where Y=0 in baseline to isolate Y-axis sensitivity.

3.3 Interpretation

The ΔZYXTS = +0.6367 demonstrates that essence injection substantially improves governance-aligned response quality. The X and T axes showed maximum improvement (+1.0 each), confirming that MoCKA essence eliminates vague language (X) and forces procedural completion (T). The S-axis partial improvement (+0.35) reflects that response length alone does not fully capture communicative efficiency — a richer S metric is proposed in Section 4.

4. Limitations and Future Work

Single-session experiment: Conditions A and B were evaluated within one session, not across independent sessions. Cross-session replication is required.

Y-axis ceiling effect: The Y-axis scored 1.0 in both conditions, indicating the test question inherently elicits re-verification language. Y-sensitive test cases must be designed.

S-axis operationalization: Current S scoring uses word count as a proxy for communicative efficiency. A richer metric incorporating information density is needed.

Single evaluator: Scores were computed by a single automated script. Inter-rater reliability with human evaluators has not been established.

5. Conclusion

XYZ+ST provides a unified five-dimensional framework for evaluating AI governance systems at both the institutional level (XYZ) and the individual response level (ST). Empirical validation demonstrates ΔZYXTS = +0.6367 between baseline and essence-injected conditions, confirming framework sensitivity. The framework is implemented in the MoCKA system and is available for replication.

The central proposition confirmed by this experiment:

"Governance structure controls reproducibility; response quality improves as a by-product."

References

[1] Kimura, M. (2026). Governing AI by Evidence, Not Faith: A Cybernetic Architecture for Auditable Multi-Model Orchestration. Zenodo. DOI: 10.5281/zenodo.19503666

[2] NIST. (2023). Artificial Intelligence Risk Management Framework (AI RMF 1.0).

[3] Wiener, N. (1948). Cybernetics: Control and Communication in the Animal and the Machine. MIT Press.